QP CODE: 25900147

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

SECOND SEMESTER MGU-UGP (HONOURS)

REGULAR EXAMINATION APRIL 2025

Second Semester

Discipline Specific Core Course–MG2DSCNBC100

BIOCHEMISTRY OF ENERGY TRANSFORMATION

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 50

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Interest (I),Appreciation(Ap),and Skill(S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple ChoiceQuestions
Answer all questions
Each question carries 1 mark

1. Option B – Co-Transporters *(1 mark)*
2. Complex I II III *(1 mark)*
3. Translation *(1 mark)*
4. Lock and Key model *(1 mark)*
5. Inner mitochondrial membrane *(1 mark)*
6. Biocatalyst *(1 mark)*

Part B
Short Answer Type Questions.
Answer any twoquestions.
Each question carries 2 marks.

7. Primary Active Transport – (½ mark for each point)

- Also called Direct Active Transport.
- Uses energy from ATP to move molecules from low to high concentration.
- Uses special proteins called ATPases or pumps in the cell membrane.
- Example: Sodium-Potassium Pump (Na^+/K^+ ATPase)

8.Cycle diagram *(1 mark)*

ATP + H20 → ADP + Pi+ 7.3 Kcal *(1 mark)*

9. Redox Potential – (1 mark for each point)

- The oxidation-reduction potential or, simply, redox potential, is a quantitative measure of the tendency of a redox pair to lose or gain electrons.
- The redox pairs are assigned specific standard redox potential (E0 volts) at pH 7.0 and 25°C.

10. mRNA – messenger RNA ***(1 mark)***

It carries genetic instructions from DNA to ribosomes ***(1 mark)***

Part C

Short Essay Type Questions.
Answer any fourquestions.
Each question carries 5 marks.

11. Osmosis –

4 marks for Explanation

1 mark for diagram

- Osmosis is the diffusion of water through a semi-permeable membrane along its concentration gradient.
- Water moves from an area of higher concentration to an area of lower concentration.
- The movement is inversely proportional to solute concentration.
- Biological membranes are semi-permeable (permeable to water and small uncharged solutes but not to large organic solutes).
- Membrane permeability depends on size, solubility, charge, and chemical properties.
- Diffusion transports various materials, while osmosis specifically transports water.
- The membrane restricts the diffusion of solutes in water.
- Aquaporins are proteins that help water move across membranes.
- Aquaporins are especially important in red blood cells and kidney tubules.

12. Oxidation, Reduction and Redox Potential –

Oxidation: It is the process in which an atom, ion, or molecule loses electrons. This loss of electrons increases the oxidation state of the substance. In simple terms, oxidation is the "loss of electrons".
Eg: - $Fe \rightarrow Fe2^{++}2e^{-}$**(Definition - 1 mark, Example ½ mark)**

Reduction: It is the process in which an atom, ion, or molecule gains electron. This gain of electrons decreases the oxidation state of the substance. In simple terms, reduction is the "gain of electrons".
Eg: -$Cu^{2+}+2e^{-} \rightarrow Cu$**(Definition - 1 mark, Example ½ mark)**

Redox Potential: - The oxidation-reduction potential or, simply, redox potential, is a quantitative measure of the tendency of a redox pair to lose or gain electrons. The redox pairs are assigned specific standard redox potential (E0 volts) at pH 7.0 and 25°C. **(Definition - 2 marks)**

13. Gene mapping- process of determining the location of genes on a chromosome & the distance between them. **(1 mark)**

Genetic mapping- Examines family history **(1 mark) (example ½ mark)**

Physical mapping – Exact distance between genes in DNA **(1 mark)(example ½ mark)**

Comparitive mapping- compares genes with between different species **(1 mark)**

14. Passive Transport –

Definition - 2 marks

Types – 3 marks for each type

- No energy required; occurs down the concentration gradient (high to low concentration).
- Naturally occurring process; cell does not use energy.
- Main types: Simple diffusion, Facilitated diffusion, and Osmosis.

Simple Diffusion

- Random movement of particles from high to low concentration.
- Continues until equilibrium (even distribution) is reached.

Facilitated Diffusion

- Movement of ions or polar molecules with the help of membrane proteins.
- Uses transmembrane integral proteins to allow passage.
- No energy used; relies on concentration gradient.
- Proteins protect molecules from the membrane's hydrophobic core.

Osmosis

- Diffusion of water through a semi-permeable membrane.

- Water moves from higher water concentration to lower water concentration.
- Movement is inversely related to solute concentration.

15. Stages of Oxidation –

Stages – 2 mark

Explanation - 3 marks

First Stage – Primary Metabolism

- Digestion in the gastrointestinal tract breaks macromolecules into small units.
- Example: Proteins → Amino acids.

• Second Stage – Secondary / Intermediary Metabolism

- Absorbed products are further catabolised and oxidised to CO_2.
- Occurs mainly in mitochondria via the citric acid cycle.
- Generates reducing equivalents: NADH and $FADH_2$.

• Third Stage – Tertiary Metabolism / Cellular Respiration

- NADH and $FADH_2$ enter the electron transport chain (ETC).
- Energy is released and used for body's synthetic activities.
- Phototrophs use light energy (e.g., plants); chemotrophs use energy from fuel oxidation.

16. DNA repair-

Direct repair- directly fixes the damaged DNA without cutting or replacing any part *(1 mark)*

excision repair- Base excision & nucleotide excision*(2 marks)*

mismatch repair- corrects the incorrect pairs. *(1 mark)*

recombination repair- Uses a healthy DNA strand as a template to fill in the missing part. *(1 mark)*

D

Essay Type Questions.

Answer any twoquestions.
Each question carries 10 marks.

17. Types of endocytosis –

Definition – 2 marks

Types- 2 marks

Explanation- 4 marks

Diagram- 2 marks

Endocytosis:

- Cell takes in large particles/molecules via vesicle formation.
- Two types: Pinocytosis and Phagocytosis.

Pinocytosis (Cell Drinking)

- Takes in liquids and dissolved substances.
- Non-specific uptake via small vesicles.

Phagocytosis (Cell Eating)

- Engulfs large solid particles like bacteria.
- Specific process using pseudopods and lysosomes.

18. High energy compounds and classification –

Definition of high energy compounds – 3 marks

Classification- 5 marks (1 mark each)

Examples – Examples and value 2 marks

- Also called energy-rich compounds.
- Release ≥ 7 kcal/mol on hydrolysis (e.g., ATP releases −7.3 kcal/mol).
- Low-energy compounds release < 7 kcal/mol.
- Most contain phosphate groups (except acetyl CoA).

- Hydrolysis yields more energy than ATP.

Classification (5 Types)

1. Pyrophosphates – e.g. ATP (−7.3 kcal/mol), has phosphoanhydride bonds.
2. Acyl phosphates – e.g. 1,3-BPG (−11.8 kcal/mol), carboxylic acid + phosphate.
3. Enol phosphates – e.g. PEP (−14.8 kcal/mol), phosphate linked to enol group.
4. Thioesters – e.g. Acetyl CoA (−7.7 kcal/mol), contains thioester bond.
5. Phosphagens – e.g. Phosphocreatine (−10.3 kcal/mol), guanidine phosphate bond.

19. Translation- Biosynthesis of protein *(1 mark)*

I Activation of amino acids *(1 mark)*

II Protein synthesis Proper

- Initiation *(2 marks)*
- Elongation *(2 marks)*
- Termination *(2 marks)*

III Post translational Modifications *(2 marks)*

20. Recombinant DNA Technology *(1 mark)*

Isolating DNA *(1 mark)*

Cutting the DNA *(1 mark)*

Preparing the Carrier (Vector DNA) *(1 mark)*

Inserting the Gene *(1 mark)*

Introducing the Recombinant DNA into the host cell *(2 marks)*

Producing copies *(1 mark)*

Extracting and Using the copies *(2 marks)*